MIL-R-22527(Wep)

15 October 1963

MILITARY SPECIFICATION

RELAY, ARMATURE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification describes the requirements for the procurement of ten types of general purpose, hermetically sealed, plug-in armature relays for use in shipboard fire control equipment. Armature relays close and open electrical contacts by means of an electromagnetically operated armature. Relay types in this specification differ with respect to their operating voltage requirements, sensitivity, transfer times, number of switching circuits, and contact ratings.

1.2 Classification. - The armature relays shall be of the following types:

Type I	BUWEPS Drawing No. 1884956,	Armature relay, miniature, four-pole, double-throw, direct-current
Type II	BUWEPS Drawing No. 1677276,	Armature relay, miniature, low-level, six-pole, double-throw, direct-current
Type III	BUWEPS Drawing No. 1884891,	Armature relay, minature, double-pole, double-throw, alternating-current
Type IV	BUWEPS Drawing No. 1677373,	Armature relay, miniature, double-pole, double-throw, direct-current
Type V	BUWEPS Drawing No. 1884896,	Armature relay, miniature, double-pole, double-throw, direct-current
Type VI	BUWEPS Drawing No. 1677371,	Armature relay, miniature, sensitive, double-pole, double-throw, direct-current
Type VII	BUWEPS Drawing No. 1884957,	Armature relay, miniature, sensitive, double-pole, double-throw, direct-current

FSC 1290

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Type VIII	BUWEPS Drawing No. 1884087,	Armature relay, miniature, slow-operate, double-pole, single-throw, direct-current
Type IX	BUWEPS Drawing No. 1677337,	Armature relay, miniature, six-pole, double-throw, alternating-current
Type X	BUWEPS Drawing No. 2038194	Armature relay, sub-miniature, low-level, double-pole, double-throw, direct-current

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATION

FEDERAL

UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Setup, Paperboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification
MIL-L-10547	Liners, Case, Waterproof

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of
MIL-W-21927	Weapons, Handling and Preparation for Delivery of, General Requirements for
MIL-STD-106	Mathematical Symbols
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUWEPS Dwg 1677276	Relay, Armature Miniature Low Level
BUWEPS Dwg 1677337	Relay, Armature, Miniature
BUWEPS Dwg 1677371	Relay, Armature Sensitive, D.C.
BUWEPS Dwg 1677373	Relay, Armature, Miniature
BUWEPS Dwg 1884891	Relay, Armature, Miniature
BUWEPS Dwg 1884896	Relay, Armature, Miniature
BUWEPS Dwg 1884956	Relay, Armature
BUWEPS Dwg 1884957	Relay, Armature Sensitive, D.C.
BUWEPS Dwg 1885087	Relay, Armature, Slow Operate
BUWEPS Dwg 2038194	Relay, Armature

PUBLICATIONS

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BUREAU OF NAVAL WEAPONS

NAVWEPS OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the armature relay shall be manufactured by the methods and procedures proposed for the production lot. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the production methods are capable of yielding items that comply with the technical requirements of the contract. See 4.1.1.

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on the opposite end should have the NSN, quantity, and

3.1.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the armature relay shall conform to the detailed requirements given on the applicable BUWEPS drawing, in the specifications listed thereon, and in Specification MIL-F-18870.

3.1.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for the armature relay shall be in accordance with the applicable BUWEPS drawing, and with the applicable specifications and publications listed thereon.

3.1.4 Moisture, fungus, and corrosion. - The armature relay shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the applicable BUWEPS drawing, and in the specifications listed therein.

3.1.5 Marking. - Marking of assemblies shall be in accordance with the applicable BUWEPS drawing.

3.1.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.1.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.8 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability and the reasons for the suggested

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deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.9 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - Definitions applicable to this specification and associated drawings are as follows:

3.2.1 Switching circuit. - A switching circuit is a movable contact plus its mating contact or contacts.

3.2.2 Pull-in voltage. - The pull-in voltage of a relay is the voltage across the coil which will cause the contacts of the relay to assume the energized position as the coil voltage is slowly increased from zero.

3.2.3 Pull-in current. - The pull-in current of a relay is the current through the coil which will cause the contacts of the relay to assume the energized position as the coil current is slowly increased from zero.

3.2.4 Operate current. - Same as pull-in current.

3.2.5 Hold voltage. - The hold voltage of a relay is the minimum voltage across the coil which will maintain the contacts of an energized relay in the energized position.

3.2.6 Hold current. - The hold current of a relay is the minumum current through the coil which will maintain the contacts of an energized relay in the energized position.

3.2.7 Drop-out voltage. - The drop-out voltage of a relay is the maximum voltage across the coil which will allow relay contacts that are initially in their energized position to assume their unenergized position.

3.2.8 Drop-out current. - The drop-out current of a relay is the maximum current through the coil which will allow relay contacts that are initially in their energized position to assume their unenergized position.

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3.2.9 Release current. - Same as drop-out current.

3.2.10 Contact chatter. - Contact chatter is the condition of sustained rapid opening and closing of relay contacts caused by mechanical shock, vibration, or other effects.

3.2.11 Contact bounce. - Contact bounce is the transient opening and closing of the contacts of a relay as the contacts are closed by energizing or de-energizing the coil.

3.2.12 Contact bounce time. - Contact bounce time is the time interval between the first instant that electrical continuity is made between contacts and the instant that unbroken continuity is established.

3.2.13 Operate time. - The operate time of a relay is the time required (including contact bounce time) from the instant power is applied to the relay coil, for the relay contacts, initially in their unenergized position, to assume their energized position.

3.2.14 Transfer time. - The transfer time of a double-throw relay is the time required (including contact bounce time) for the movable contact of the relay to shift from one stationary contact to the other stationary contact.

3.2.15 Release time. - The release time of a relay is the time required (including contact bounce time) from the instant power is removed from the relay coil, for the relay contacts, initially in their energized position, to assume their unenergized position.

3.2.16 Coil temperature rise. - The coil temperature rise corresponding to the coil resistance increasing from R_C to R_H is given by:

$$(234.5 + T_C)\frac{R_H - R_C}{R_C}$$

where R_H = resistance of coil at temperature T_H,
R_C = resistance of coil at temperature T_C,
T_H = temperature of coil when energized,
T_C = room ambient temperature (see 4.4.2.1),

and all temperatures are in degrees centigrade.

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(USED WHEN REQUIRED)

Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

3.2.17 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which present an opportunity for the growth of fungi.

3.3 Performance requirements and product characteristics. - The armature relay shall meet the following performance requirements and product characteristics:

3.3.1 Dielectric strength and insulation resistance.

3.3.1.1 Dielectric strength. - When tested in accordance with 4.4.4.1.1, the armature relay shall withstand the dielectric test voltage specified in 4.4.4.1.1 without arcing, breakdown, or damage, and the dielectric leakage current shall not exceed 1 milliampere. See 4.4.4.1.1.

3.3.1.2 Insulation resistance. - When measured in accordance with 4.4.4.1.2, the insulation resistance between mutually insulated terminals, and between insulated terminals and the case shall be 1000 megohms minimum unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.1.2.

3.3.2 Coil resistance and current.

3.3.2.1 Direct-current relays. - When measured in accordance with 4.4.4.2.1, the coil resistance of armature relay types I, II, IV, V, VI, VII, VIII, and X shall meet the requirements of the applicable BUWEPS drawing. See 4.4.4.2.1.

3.3.2.2 Alternating-current relays. - When measured in accordance with 4.4.4.2.2, the coil current of armature relay types III and IX shall meet the requirements of the applicable BUWEPS drawing. See 4.4.4.2.2.

3.3.3 Contact resistance.

3.3.3.1 Non-low-level relays. - When measured in accordance with 4.4.4.3.1, the contact resistance between any pair of mating contacts of armature relay types I, III, IV, V, VI, VII, VIII, and IX shall not exceed 0.05 ohm unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.3.1.

3.3.3.2 Low-level relays. - When measured during run-in cycling in accordance with 4.4.4.3.2, the contact resistance between mating pairs of contacts of armature relay types II and X shall not exceed 800 ohms more than two times. See 4.4.4.3.2.

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3.3.4 Pull-in voltage and current.

3.3.4.1 Voltage relays.

3.3.4.1.1 Direct-current voltage relays. - When checked in accordance with 4.4.4.4, armature relay types I, II, IV, V, VIII, and X shall pull in and all switching circuits shall make positive contact or opening, as applicable, at a coil voltage not greater than the following, as applicable, for ambient temperature not greater than 35 degrees centigrade (95 degrees Fahrenheit):

(a) The pull-in voltage specified on the applicable BUWEPS drawing if the pull-in voltage is specified thereon,
(b) 57 percent of rated coil voltage as specified on the applicable BUWEPS drawing if the pull-in voltage is not specified thereon.

See 4.4.4.4.

3.3.4.1.2 Alternating-current voltage relays. - When checked in accordance with 4.4.4.4, armature relay types III and IX shall pull in and all switching circuits shall make positive contact or opening, as applicable, at a coil voltage not greater than the following, as applicable, for ambient temperature not greater than 35 degrees centigrade (95 degrees Fahrenheit):

(a) The pull-in voltage specified on the applicable BUWEPS drawing if the pull-in voltage is specified thereon,
(b) 77 percent of rated coil voltage as specified on the applicable BUWEPS drawing if the pull-in voltage is not specified thereon.

See 4.4.4.4.

3.3.4.2 Current relays. - When checked in accordance with 4.4.4.4, armature relay types VI and VII shall pull in and all switching circuits shall make positive contact or opening, as applicable, at a coil current not greater than the operate current specified on the applicable BUWEPS drawing. See 4.4.4.4.

3.3.5 Hold voltage. - When checked in accordance with 4.4.4.5, armature relay types I, II, III, IV, V, VIII, IX, and X shall remain pulled in with all switching circuits maintaining positive contact or open circuit, as applicable, when the coil voltage is decreased from a value greater than the pull-in voltage to 50 percent of rated coil voltage as specified on the applicable BUWEPS drawing unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.5.

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3.3.6 Drop-out voltage and current.

3.3.6.1 Voltage relays.

3.3.6.1.1 Direct-current voltage relays. - When checked in accordance with 4.4.4.6, armature relay types I, II, IV, V, VIII, and IX shall drop out and all switching circuits shall open or make positive contact, as applicable, when the coil voltage is decreased from a value greater than pull-in voltage to not less than 1.0 volt unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.6.

3.3.6.1.2 Alternating-current voltage relays. - When checked in accordance with 4.4.4.6, armature relay types III and IX shall drop out and all switching circuits shall open or make positive contact, as applicable, when the coil voltage is decreased from a value greater than pull-in voltage to not less than 9 volts root-mean-square (r.m.s.) unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.6.

3.3.6.2 Current relays. - When checked in accordance with 4.4.4.6, armature relay types VI and VII shall drop out and all switching circuits shall open or make positive contact, as applicable, when the coil current is decreased from a value greater than pull-in current to not less than the release current specified on the applicable BUWEPS drawing. See 4.4.4.6.

Table I. Switching Transient Times (in Milliseconds)

Armature Relay Type	I	II, V, VI, VII	III	IV	VIII	IX	X
Contact Bounce Time (Max)	3	3	6	3	3	3	2
Transfer Time (Max)	6	6	6	6	6	6	3
Operate Time (Min)	-	-	-	-	25	-	-
Operate Time (Max)	20	15	20	8	150	27	10
Release Time (Max)	10	10	20	5	10	27	6

3.3.7 Contact bounce time. - When measured in accordance with 4.4.4.7, the contact bounce time shall meet the requirements of table I unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.7.

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3.3.8 Transfer time. - When measured in accordance with 4.4.4.8, the transfer time shall meet the requirements of table I unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.8.

3.3.9 Operate time. - When measured in accordance with 4.4.4.9, the operate time shall meet the requirements of table I unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.9.

3.3.10 Release time. - When measured in accordance with 4.4.4.10, the release time shall meet the requirements of table I unless otherwise specified on the applicable BUWEPS drawing. See 4.4.4.10.

3.4 Environmental requirements. - The armature relay shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1.1 Vibration, transportation, storage, and handling. - The armature relay shall withstand the tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.1.2 Vibration, operating condition. - The armature relay shall withstand the type I vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.1.2.

3.4.2 Shock

3.4.2.1 Shock, transportation, storage, and handling. - The armature relay shall withstand the tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.2.2 Shock, non-operating condition. - The armature relay shall withstand the type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.2.2.

3.4.3 Temperature

3.4.3.1 Temperature, operating condition. - The armature relay shall withstand the operational ambient temperatures between the applicable limits given in table II in accordance with Specification MIL-F-18870, shall function during and after temperature tests (see notes 1 and 2 below), and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.1.

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3.4.3.2 Temperature, non-operating condition. - The armature relay shall withstand storage temperatures between the applicable limits given in table II in accordance with Condition II of OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.2.

Table II. Operating and Non-Operating Temperature Ranges

Armature Relay Type	Operating Condition	Non-Operating Condition
I, III	-67°F. to 257°F. (-55°C. to 125°C.)	-85°F. to 257°F. (-65°C. to 125°C.)
II, IV, V, VIII, IX, X	-67°F. to 185°F. (-55°C. to 85°C.)	-85°F. to 185°F. (-65°C. to 85°C.
VI, VII	-67°F. to 212°F. (-55°C. to 100°C.)	-85°F. to 212°F. (-65°C. to 100°C.)

Note 1. Pull-in voltage. - For ambient temperature greater than 95 degrees Fahrenheit (35 degrees centigrade), the required minimum pull-in voltage shall be as follows (see 3.3.4.1).

(a) The pull-in voltage specified on the applicable BUWEPS drawing if the pull-in voltage is specified thereon,

(b) The percentage given in table III of rated coil voltage as specified on the applicable BUWEPS drawing if the pull-in voltage is not specified thereon.

Table III. Pull-in Voltage as Percentage of Rated Coil Voltage

Armature Relay Type	Ambient Temperature	
	95°F. to 185°F. (35°C. to 85°C.)	185°F. to 257°F. (85°C. to 125°C)
D-C Voltage Relays	68%	76%
A-C Voltage Relays	85%	92%

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Note 2. Insulation resistance. - For rated voltage or 120 percent of rated pull-in current, as applicable, as specified on the applicable BUWEPS drawing applied to the armature relay and an ambient temperature of 257 degrees Fahrenheit (125 degrees centigrade), the required insulation resistance after temperature stabilization has been attained shall be 100 megohms minimum.

3.4.4 Humidity. - The armature relay shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.4.

3.4.5 Tropical environment. - The armature relay shall withstand the tropical environmental conditions specified in Specification MIL-E-5272 under "Fungus Resistance Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification. See 4.5.5.

3.4.6 Coil temperature rise. - When measured in accordance with 4.5.6, the temperature of the coil shall not exceed 200 degrees centigrade (392 degrees Fahrenheit). See 4.5.6.

3.4.7 Coil overload. - After the armature relay has been subjected to the coil overload test described in 4.4.4.10, the requirements of 3.3.1 (coil to case only), 3.3.2, 3.3.4, and 3.3.6 shall be met. See 4.5.7.

3.4.8 Contact overload. - After the armature relay has been subjected to the contact overload test described in 4.5.8, the contact resistance between any pair of mating contacts shall not exceed 0.05 ohm. See 4.5.8.

3.4.9 Operating life.

3.4.9.1 Non-low-level voltage relays. - Armature relay types I, III, IV, V, VIII, and IX shall withstand the operating life test described in 4.5.9.1, and thereafter shall meet all the performance requirements and product characteristics of this specification except that

(a) A contact resistance not exceeding 0.10 ohm shall be acceptable unless otherwise specified on the applicable BUWEPS drawing,

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(b) The high-potential voltage for checking dielectric strength after completion of the life test shall be 500 volts r.m.s. unless otherwise specified on the applicable BUWEPS drawing.

See 4.5.9.1

3.4.9.2 Low-level voltage relays. - When measured during operation in accordance with 4.5.9.2(a), the contact resistance between any pair of mating contacts shall not exceed 800 ohms at 8 microamperes more than 4 times for armature relay types II and X. Armature relay types II and X shall withstand the operating life test described in 4.5.9.2, and thereafter shall meet all the performance requirements and product characteristics of this specification except that

(a) The requirements of 3.3.3.2 shall not be applicable,
(b) A contact resistance not greater than 0.10 ohm at 100 milliamperes shall be acceptable unless otherwise specified on the applicable BUWEPS drawing,
(c) The high-potential voltage for checking dielectric strength after completion of the life test shall be 500 volts r.m.s. unless specified on the applicable BUWEPS drawing.

See 4.5.9.2.

3.4.9.3 Current relays. - Armature relay types VI and VII shall withstand the operating life test described in 4.5.9.3, and thereafter shall meet all the performance requirements and product characteristics of this specification except that

(a) A contact resistance not exceeding 0.10 ohm shall be acceptable unless otherwise specified on the applicable BUWEPS drawing,
(b) The high-potential voltage for checking dielectric strength following completion of the life test shall be 250 volts r.m.s. unless otherwise specified on the applicable BUWEPS drawing.

See 4.5.9.3.

3.4.10 Hermetic seal. - After the armature relay has been subjected to the hermetic seal test of 4.5.10, there shall be no evidence of leakage when the relay housing is opened.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.6.1 In addition to the examination and tests performed on the pre-production samples, the workmanship exhibited therein will be evaluated to determine acceptability. The approved standard of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three armature relays shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic Production Samples. - The tests of 4.4 and 4.5 shall be performed in steps as specified below. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

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Step 2. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty-five (25).

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.

4.2 Classification of tests. - The inspection and testing of the armature relay shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each armature relay submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.1 and 4.1.2. These tests are detailed in 4.5. Acceptance shall be based on no defects in a sample.

4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the armature relay. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 High-potential tester. - A high-potential tester capable of applying up to 1000 volts r.m.s., 60 cycles per second, which cuts out or indicates when the dielectric leakage current exceeds 1 milliampere.

4.3.2 Insulation resistance tester. - An insulation resistance tester capable of applying 100 volts direct current, and measuring resistance up to 2000 megohms.

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4.3.3 Power sources.

4.3.3.1 Low-voltage direct-current power supply.- A direct-current power supply having the following characteristics:

(a) Voltage.- 2 volts minimum, 6 volts maximum, adjustable or non-adjustable,
(b) Current.- 6 amperes minimum.

4.3.3.2 Non-adjustable direct-current power supply.- A direct-current power supply having the following characteristics:

(a) Voltage.- 27.25 volts plus or minus 1 volt,
(b) Current.- 18 amperes minimum.

4.3.3.3 Adjustable direct-current power supply.- A direct-current power supply having the following characteristics:

(a) Voltage.- Continuously adjustable from 0.5 volt maximum to 60 volts minimum,
(b) Current.- 300 milliamperes minimum at any output voltage within the rated voltage range.

4.3.3.4 400-cycle power supply.- A 400-cycle power supply having the following characteristics:

(a) Voltage.- Continuously adjustable from 5 volts maximum to 135 volts minimum,
(b) Current.- 18 amperes minimum,
(c) Frequency.- 400 plus or minus 20 cycles per second,
(d) Waveform.- Sinusoidal, with harmonic content not greater than 5 percent of the voltage of fundamental frequency.

4.3.3.5 60-cycle equipment power supply.- A 60-cycle power source capable of supplying the current required by any 60-cycle test equipment used in conducting the acceptance tests set forth in this specification, and having the following characteristics:

(a) Voltage.- 115 plus or minus 2 volts,
(b) Frequency.- 60 plus or minus 3 cycles per second,
(c) Waveform.- Sinusoidal, with harmonic content not greater than 5 percent of the voltage of fundamental frequency.

4.3.4 Meters.

4.3.4.1 Direct-current milliammeter.- A multi-range, direct-current milliammeter, 1, 3, 10, 30, 100, and 300 milliamperes full scale, having

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an impedance not greater than 0.20 ohm, and an accuracy of plus or minus 1 percent of full scale. Two such milliammeters are required.

4.3.4.2 Alternating-current milliammeter.- An alternating-current milliammeter, 50 milliamperes full scale, having an impedance not greater than 4 ohms, and an accuracy of plus or minus 1 percent of full scale.

4.3.4.3 Direct-current ammeter.- A direct-current ammeter, 5 amperes full scale, having an accuracy of plus or minus 2 percent of full scale.

4.3.4.4 Direct-current millivoltmeter.- A direct-current millivoltmeter, 250 millivolts full scale, having a resistance not less than 1000 ohms, and an accuracy of plus or minus 2 percent of full scale.

4.3.4.5 Direct-current voltmeter.- A multi-range, direct-current voltmeter, 1, 3, 10, 30, and 100 volts full scale, having a resistance of at least 10,000 ohms per volt, and an accuracy of plus or minus 2 percent of full scale.

4.3.4.6 Alternating-current voltmeter.- A multi-range, alternating-current voltmeter, 10, 30, 100, and 300 volts full scale, having an input impedance of at least 10,000 ohms per volt, and an accuracy of plus or minus 1 percent of full scale.

4.3.5 Decade resistor.- A decade resistor adjustable to 100,000 ohms in steps of 1 ohm, having a power rating of 5 watts minimum, and an accuracy of plus or minus 5 percent.

4.3.6 Wheatstone bridge.- A direct-current resistance bridge capable of measuring resistance from 10 to 30,000 ohms with an accuracy of plus or minus 0.5 percent.

4.3.7 Oscilloscope.- A direct-coupled oscilloscope having the following characteristics:

(a) Phosphor.- Long persistence
(b) Sweep speed.- Adjustable so that pulses 1 to 300 milliseconds in length can be conveniently displayed,
(c) Sweep speed accuracy.- A sweep speed or timing wave accuracy of plus or minus 1 percent,
(d) Sweep triggering.- The sweep generator shall be capable of being externally triggered,
(e) Rise time.- 10 microseconds maximum,
(f) Input impedance, each axis.- 0.5 megohm minimum.

4.3.8 Low-level run-in tester.- A tester, with two counters, which

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performs the following operations:

(a) Coil. - Interrupts coil energizing voltage (rated coil voltage as specified on the applicable BUWEPS drawing) at a rate not less than 60 and not greater than 180 cycles per minute with on time equal to off time, and actuates one counter each time a coil voltage on-off cycle is completed,

(b) Contact current. - Supplies a 400 to 800 cycle per second current of 8 microamperes plus or minus 10 percent to all pairs of mating contacts with a corresponding open contact voltage not greater than 250 millivolts r.m.s. from a purely resistive power source,

(c) Contact resistance. - Measures the contact resistance of all pairs of mating contacts on each half of the coil voltage cycle, and actuates the other counter each time one or more contact pairs have a resistance greater than 800 ohms. (In lieu of measuring the resistance of each contact pair individually, it is permissible to measure the sum of the contact-pair resistances on each half of the coil voltage cycle. If this is done, the counter shall be actuated each time the sum of the resistances exceeds 800 ohms.)

4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the armature relay is in compliance with the requirements of 3.1, 3.3, 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

4.4.2 Test conditions. - Unless otherwise specified, the armature relay shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are spe-

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cified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure.- The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Dielectric strength and insulation resistance.

4.4.4.1.1 Dielectric strength.- Using the high-potential tester of 4.3.1, the following voltages shall be applied for 5 seconds. The high-potential tester shall be observed for indication of dielectric leakage current during application of high-potential voltage, and the relay shall be examined for any evidence of damage, arcing, or breakdown after application of high-potential voltage to determine compliance with the requirements of 3.3.1.1.

4.4.4.1.1.1 Voltage relays (Armature Relay types I, II, III, IV, V, VIII, IX, and X).

1000 Volts r.m.s.

(a) Between each switching circuit and all other switching circuits,
(b) Between all switching circuits and the coil (relay types III and IX excluded),
(c) Between all switching circuits and the case.

500 Volts r.m.s.

(a) Between open contacts within all switching circuits in both the energized and non-energized conditions,
(b) Between the coil and the case (relay types III and IX excluded).

4.4.4.1.1.2 Current relays (Armature Relay types VI and VII).

500 Volts r.m.s.

(a) Between each switching circuit and all other switching circuits,
(b) Between all switching circuits and the coil,
(c) Between all switching circuits and the case,
(d) Between the coil and the case.

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300 Volts r.m.s.

(a) Between open contacts within all switching circuits in both the energized and non-energized conditions.

4.4.4.1.2 Insulation resistance.- Measurement of insulation resistance shall be made between the following points using the insulation resistance tester of 4.3.2 to determine compliance with the requirements of 3.3.1.2. No connection shall be made to the coil of Armature Relay types III and IX.

(a) Between mutually insulated terminals,
(b) Between open contacts within all switching circuits in both the energised and unenergised conditions,
(c) Between the coil and the case,
(d) Between contacts and the case.

4.4.4.2 Coil resistance and current.

4.4.4.2.1 Direct-current relays.- The coil resistance of Armature Relay types I, II, IV, V, VI, VII, VIII, and X shall be measured using the Wheatstone bridge of 4.3.6 to determine compliance with the requirements of 3.3.2.1.

4.4.4.2.2 Alternating-current relays.- The coil current of Armature Relay types III and IX shall be measured by connecting in series the relay, the direct-current milliammeter of 4.3.4.1, and the power supply of 4.3.3.4 - adjusted to rated coil voltage as specified on the applicable BUORD drawing, and observing the reading of the milliammeter to determine compliance with the requirements of 3.3.2.2.

4.4.4.3 Contact resistance.

4.4.4.3.1 Non low-level relays.- The circuit of figure 1 shall be used to measure the voltage drop across each pair of mated contacts of Armature Relay types I, III, IV, V, VI, VII, VIII, and IX for both the energised and non-energised conditions to determine compliance with the requirements of 3.3.3.1. Contact resistance shall be computed by dividing the reading of the direct current millivoltmeter in volts by the reading of direct-current ammeter in amperes. The relay coil shall be energised by rated coil voltage as specified on the applicable BUORD drawing, using the power supply of 4.3.3.3 or 4.3.3.4 as applicable. The contact current shall be rated contact current for direct-current resistive-load operation as specified on the applicable BUORD drawing, and the open-contact voltage shall not exceed 6 volts.

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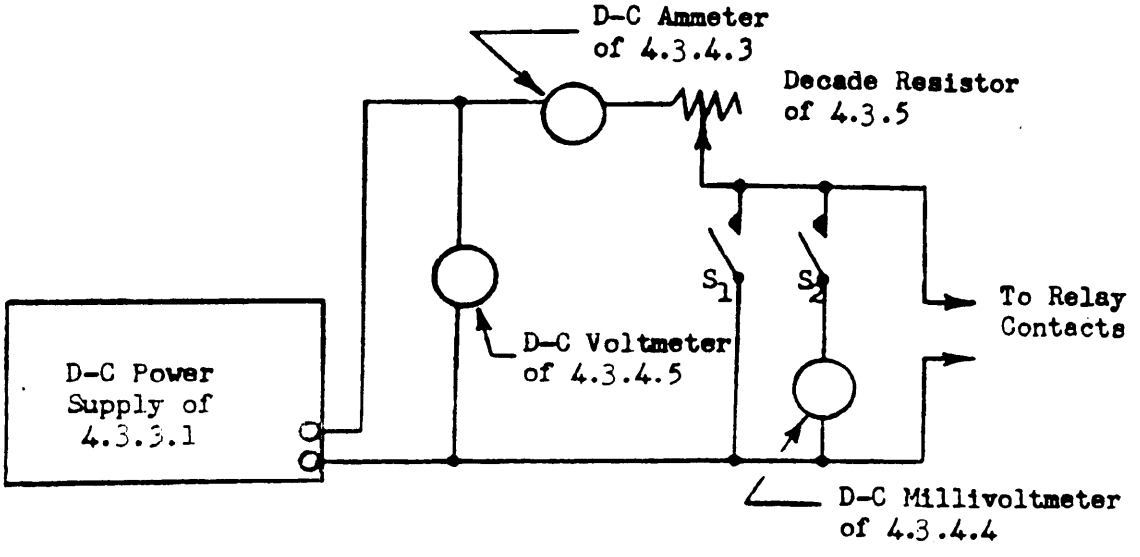


Figure 1. Circuit for Measurement of Contact Resistance

With switch S_1 closed, the output current of the power supply shall be adjusted to rated contact current of the Armature Relay to be tested before making connection to the contacts of the Armature Relay. Switch S_2, which permits operation of the direct-current millivoltmeter, shall be closed only if the direct-current ammeter indicates that rated current is flowing. To avoid damage to the direct-current millivoltmeter, the condition of the Armature Relay (energized or non-energized) shall not be changed with switch S_2 closed.

4.4.4.3.2 Low-level relays.- Armature Relay types II and X shall be subjected to 10,000 cycles of run-in cycling using the low-level run-in tester of 4.3.8. At the end of the 10,000 cycles, the reading of the counter which is actuated in response to measured contact resistance exceeding 800 ohms shall be noted to determine compliance with the requirements of 3.3.3.2.

4.4.4.4 Pull-in voltage and current.- The pull-in voltage of

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Armature Relay types I, II, III, IV, V, VIII, IX, and X, and the pull-in current of Armature Relay types VI and VII shall be measured to determine compliance with the requirements of 3.3.4.1.2, 3.3.4.1.2, or 3.3.4.2, as applicable, using the circuit of figure 2. The voltage of the power supply energizing the coil shall be increased slowly from its lowest value until the milliammeter connected to the contacts indicates an abrupt increase in current. The coil voltage or current, as applicable, at this point is the pull-in value. The coil of Armature Relay types I, II, IV, V, VI, VII, VIII, and X shall be energized by the power supply of 4.3.3.3, and the coil of Armature Relay types III and IX shall be energized by the power supply of 4.3.3.4.

Before making connection to the contacts of the Armature Relay, the output current of the power supply of 4.3.3.1 shall be adjusted to 100 milliamperes with the circuit for detecting contact current closed.

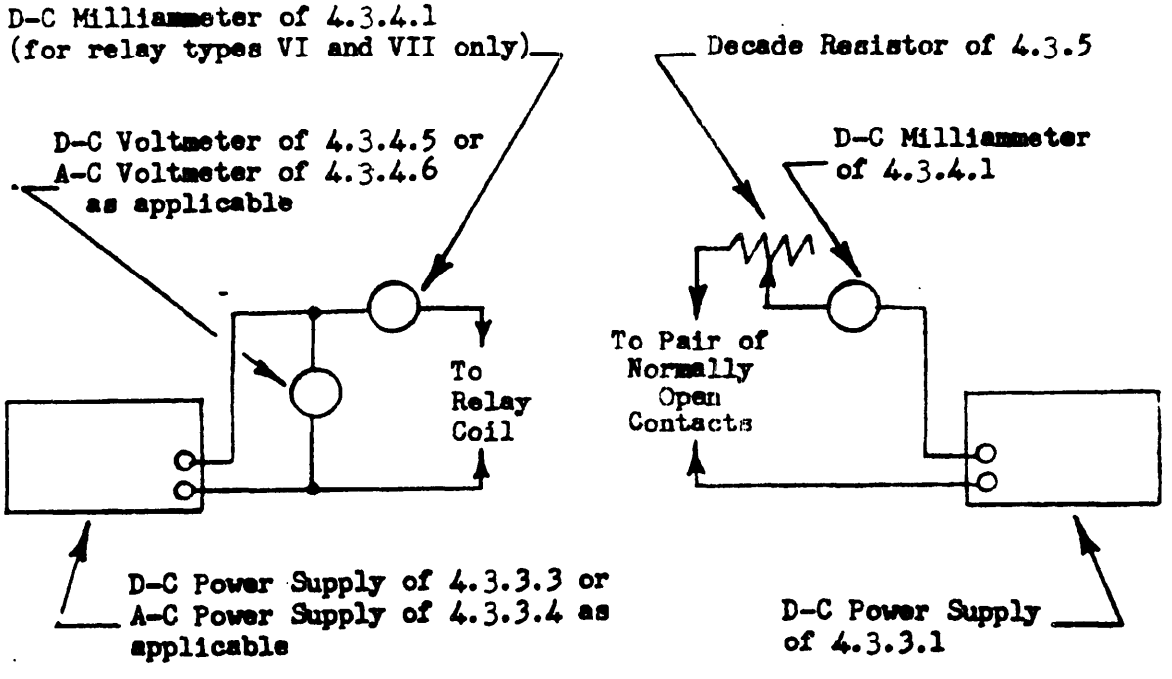


Figure 2. Circuit for Measurement of Pull-In Voltage or Current, Hold Voltage, and Drop-Out Voltage or Current

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the item with burlap or other suitable covering. With each end of

4.4.4.5 Hold voltage.- Using the circuit of figure 2, with the output current of the power supply of 4.3.3.1 as specified in 4.4.4.4, the coil of Armature Relay types I, II, III, IV, V, VIII, IX, and X shall be energized with rated coil voltage as specified on the applicable BUORD drawing. The coil voltage shall then be decreased slowly to 50 percent of this value, and the indication of the direct-current milliammeter connected to the relay contacts shall be noted to determine compliance with the requirements of 3.3.5.

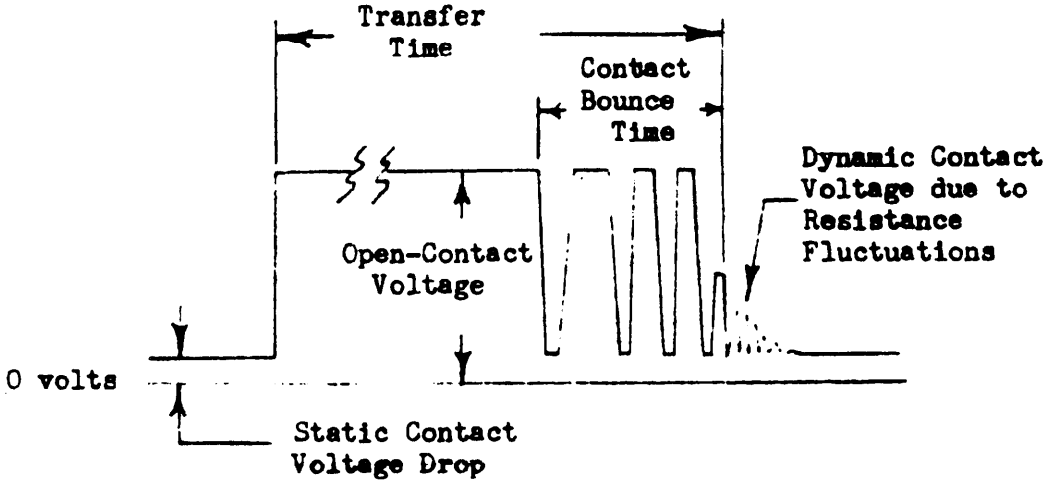
4.4.4.6 Drop-out voltage or current.- The drop-out voltage of Armature Relay types I, II, III, IV, V, VIII, IX, and X, and the drop-out current of Armature Relay types VI and VII shall be measured to determine compliance with the requirements of 3.3.6.1.1, 3.3.6.1.2, or 3.3.6.2, as applicable, using the circuit of figure 2, with the output current of the power supply of 4.3.3.2 as specified in 4.4.4.4. The voltage of the power supply energizing the coil shall be adjusted to give rated coil voltage or pull-in current, as applicable, as specified on the applicable BUORD drawing, and then decreased slowly until the reading of the direct current milliammeter connected to the contacts drops abruptly to zero. The coil voltage or current, as applicable, at this point is the drop-out value.

Table IV. Number of Switching Circuits per Relay to be Checked

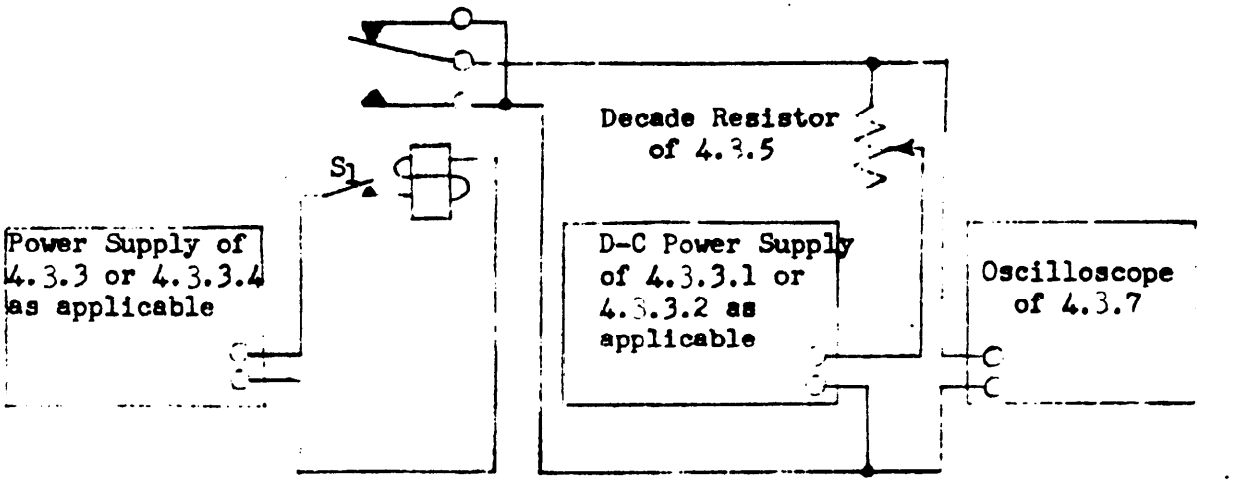
Total Number of Switching Circuits	Number of Switching Circuits, Chosen at Random, to be Checked
1	1
2	1
3	1
4	1
5	2
6	2

4.4.4.7 Contact bounce time.- Contact bounce time shall be measured as indicated in figure 3 to determine compliance with the requirements of 3.3.7. The voltage of the power supply energizing the coil shall be adjusted to give rated coil voltage or 120 percent of rated pull-in current, as applicable, as specified on the applicable BUORD drawing, with switch S_1 closed. For Armature Relay types I, III, IV, V, VI, VII, VIII, and IX, the contacts shall be energized using the power supply of 4.3.3.2, with contact current equal to rated contact current for direct-current resistive-load operation as specified on the applicable BUORD drawing. For Armature Relay types II and X, the contacts shall be energized using the power supply of 4.3.3.1, with a contact current of 100 milliamperes. The number of switching circuits per relay for which contact bounce time is measured shall be in accordance with table IV.

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Typical oscilloscope trace of test circuit



Test circuit for measurement of contact bounce time and transfer time

Notes:

1. Closing switch S_1 produces the oscilloscope trace giving bounce of normally-open contacts, and transfer time on operate.

2. Opening switch S_1 produces the oscilloscope trace giving bounce of normally-closed contacts, and transfer time on release.

Figure 3. Measurement of Contact Bounce Time and Transfer Time

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4.4.4.8 Transfer time. - Transfer time shall be measured as indicated in figure 3 to determine compliance with the requirements of 3.3.8. The test circuit power supply voltages and currents shall be as specified in 4.4.4.7, and the number of switching circuits per relay for which transfer time is measured shall be in accordance with table IV.

4.4.4.9 Operate time. - Operate time shall be measured as indicated in figure 4 to determine compliance with the requirements of 3.3.9. The test circuit power supply voltages and currents shall be as specified in 4.4.4.7, and the number of switching circuits per relay for which operate time is measured shall be in accordance with table IV.

4.4.4.10 Release time. - Release time shall be measured as indicated in figure 4 to determine compliance with the requirements of 3.3.10. The test circuit power supply voltages and currents shall be as specified in 4.4.4.7, and the number of switching circuits per relay for which release time is measured shall be in accordance with table IV.

4.4.4.11 Preservation, packing and packaging. - Preservation, packing and packaging of the relay shall conform to section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Vibration

4.5.1.1 Vibration, transportation. - The armature relay, packaged and packed as specified in section 5., shall be vibration tested as specified in Specification MIL-W-21927.

4.5.1.2 Vibration, operating condition. - The armature relay, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.4.1.2 and shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.5.2 Shock

4.5.2.1 Shock, transportation, storage and handling. - The armature relay, packaged and packed as specified in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

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the right side near the rear of the vehicle the vehicle or on

4.5.2.2 Shock, non-operating condition. - The armature relay, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in 3.4.2.2, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperature

4.5.3.1 Temperature, operating condition. - The armature relay shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3.2 Temperature, non-operating condition. - The armature relay shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.2, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.4 Humidity. - The armature relay shall be subjected to humidity tests so as to meet the requirements specified in 3.4.4, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.5 Tropical environment. - The armature relay shall be exposed to fungus tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.6 Coil temperature rise. - The armature relay shall be operated for 2 hours minimum under the following conditions, and coil temperature shall then be computed using the formula of 3.2.1.6 to determine compliance with the requirements of 3.4.6.

(a) The coil shall be energized with rated coil voltage or 120 percent of rated pull-in current as specified on the applicable BUWEPS drawing, as applicable,

(b) Each pair of mating contacts shall carry the greatest rated contact current specified on the applicable BUWEPS drawing,

(c) The armature relay shall be mounted vertically on a panel of heat insulating material in a 257-degree Fahrenheit (125-degree centigrade) ambient temperature.

The contacts shall be energized using the decade resistor of 4.3.5 and the power supply of 4.3.3.2 or 4.3.3.4, as applicable, with all mating contacts connected in series. The coil shall be energized using the power supply of 4.3.3.3 or 4.3.3.4, as applicable.

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As this test acts to increase the contact resistance for low-level currents, it shall not be performed on armature relay types II or X if subsequent measurement of contact resistance at a contact current of 100 milliamperes or less is to be made for these relays.

4.5.7 Coil overload. - The armature relay shall be operated under the following conditions, and shall then be tested in accordance with 4.4.4.1 (coil to case only), 4.4.4.2, 4.4.4.4, and 4.4.4.6 to determine compliance with the requirements of 3.4.7.

(a) The coil voltage for voltage relays (armature relay types I, II, III, IV, V, VIII, IX, and X) shall be 110 percent of rated coil voltage as specified on the applicable BUWEPS drawing, and the coil current for current relays (armature relay types VI and VII) shall be that current which would produce a dissipation of 0.5 watt in a resistance equal to 148 percent of the coil resistance specified on the applicable BUWEPS drawing.

(b) The coil shall be energized for 100 hours with the ambient temperature alternately the maximum operating temperature specified in table I for 8 hours, and room ambient temperature for 8 hours.

The coil of the voltage relays shall be energized using the power supply of 4.3.3.3 or 4.3.3.4, as applicable, and the coil of the current relays shall be energized using a constant-current generator.

4.5.8 Contact overload. - The armature relay shall be operated under the following conditions, and the contact resistance between all pairs of normally open contacts shall then be measured in accordance with 4.4.4.3.1 to determine compliance with the requirements of 3.4.8.

(a) The coil voltage for voltage relays (armature relay types I, II, III, IV, V, VIII, IX, and X) shall be rated coil voltage as specified on the applicable BUWEPS drawing, and the coil current for current relays (armature relay types VI and VII) shall be 120 percent of rated pull-in current as specified on the applicable BUWEPS drawing.

(b) Power to the coil shall be interrupted at the rate of twenty cycles per minute for a total of 100 cycles, with on time equal to off time.

(c) Each pair of mating contacts shall carry 4 times the rated current for direct-current resistive-load operation as specified on the applicable BUWEPS drawing.

The contacts shall be energized using the decade resistor of 4.3.5 and the power supply of 4.3.3.2 with all mating contacts connected in series. The coil shall be energized using the power supply of 4.3.3.3 or 4.3.3.4, as applicable.

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As this test acts to increase the contact resistance for low level currents, it shall not be performed on armature relay types II or X if subsequent measurement of contact resistance at a contact current of 100 milliamperes or less is to be made for these relays.

4.5.9 Operating life. - The armature relay shall be operated under the following conditions so as to meet the requirements specified in 3.4.9, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.9.1 Non-low-level voltage relays (armature relay types I, III, IV, V, VII, and IX).

(a) The coil voltage shall be rated coil voltage as specified on the applicable BUWEPS drawing,

(b) Power to the coil shall be interrupted at the rate of 50 to 60 cycles per minute for a total of 100,000 cycles minimum, with on time equal to off time, unless otherwise specified on the applicable drawing,

(c) Each pair of mating contacts shall carry the current specified on the applicable BUWEPS drawing or below, as applicable, for inductive direct-current operation, and the time constant of the contact circuit load impedance shall be 0.008 second, inductive,

(d) The test shall be conducted in 24-hour cycles - 16 hours of operation in accordance with 4.5.9.1(a) at an ambient temperature equal to the maximum temperature for the operating condition as specified in table II, followed by 8 hours off at room ambient temperature.

Armature Relay Type	Current for Inductive Circuit, 28 Volts D.C.
I	0.5 A.
IV	1 A.
VIII	0.5 A.

The contacts shall be energized using the decade resistor of 4.3.5, a suitable inductor, and the power supply of 4.3.3.2 with all mating contacts connected in series. The coil shall be energized using the power supply of 4.3.3.3 or 4.3.3.4, as applicable.

4.5.9.2 Low-level voltage relays (armature relay types II and X).

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(a) The relays shall be operated for 100,000 cycles using the low-level run-in tester of 4.3.8. At the end of the 100,000 cycles, the reading of counter which is actuated in response to measured contact resistance exceeding 800 ohms shall be noted to determine compliance with the 8-microampere contact resistance requirements of 3.4.9.2,

(b) The relays shall then be subjected to cycling in accordance with 4.5.9.1. At the end of this cycling, the contact resistance shall be measured in accordance with 4.4.4.3.1 - except that the contact current shall be 100 milliamperes and the milliammeter of 4.3.4.1 shall be used in place of the ammeter of 4.3.4.3, to determine compliance with the requirements of 3.4.9.2(b).

4.5.9.3 Current relays (armature relay types VI and VII).

(a) The coil current shall be 120 percent of rated pull-in current as specified on the applicable BUWEPS drawing,

(b) Power to the coil shall be interrupted at the rate of 20 to 30 cycles per minute for a total of 100,000 cycles minimum, with on time equal to off time, unless otherwise specified on the applicable BUWEPS drawing,

(c) Each pair of mating contacts shall carry the current specified on the applicable BUWEPS drawing for direct-current resistive-load operation.

(d) The test shall be conducted in 24-hour cycles - 16 hours of operation in accordance with 4.5.9.3(a) at an ambient temperature of 212 degrees Fahrenheit (100 degrees centigrade) followed by 8 hours off at room ambient temperature.

The contacts shall be energized using the decade resistor of 4.3.5 and the power supply of 4.3.3.2 with all mating contacts connected in series.

4.5.10 Hermetic seal. - The armature relay shall be immersed in a saturated solution of sodium chloride with a suitable wetting agent at an absolute pressure of 2.5 inches of mercury for 4 hours followed by 4 hours continued immersion at normal atmospheric pressure. The housing shall then be opened and inspected to determine compliance with the requirements of 3.4.10.

4.6 Waiver of tests. - Tests specified in 4.5.1.1 and 4.5.2.1 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

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4.7 Functional tests. - At the conclusion of the tests specified in 4.5.1.1 and 4.5.2.1, functional and other tests, as applicable, shall be conducted on the armature relay to determine freedom from operational malfunction and compliance with all the requirements of section 3.

4.8 Failure of the armature relay to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging. - Unit packaging of the armature relay shall be in accordance with Method IIe of Specification MIL-P-116. The unit package shall contain one (1) armature relay packaged in a paperboard box in accordance with Specification PPP-B-566 or PPP-B-676 with Kraft paper overwrap in accordance with Specification UU-P-268.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package, intermediate package (when used) and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the armature relay will meet all the requirements of section 3 after being tested in accordance with 4.5.1.1 and 4.5.2.1.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging. - Unit packaging of the armature relay shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one (1) armature relay.

5.1.3.3 Intermediate package. - Not applicable.

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5.2 Packing.

5.2.1 Level A. - In addition to the following, level A packing shall be in accordance with Specification MIL-P-17555.

5.2.1.1 Exterior containers. - Armature relays packaged as described in 5.1.1.3 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, class 2 (gross weight not to exceed 150 lbs).

5.2.1.2 Case liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

5.2.2 Level B. - In addition to the following, level B packing shall be in accordance with Specification MIL-P-17555.

5.2.2.1 Exterior containers. - Armature relays packaged as described in 5.1.1.3 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type, or PPP-B-621, class 1 (gross weight not to exceed 150 lbs) or in fiber boxes in accordance with Specification PPP-B-636, class 1 (gross weight not to exceed 70 lbs).

5.2.2.2 Case liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

5.2.3 Level C. - Armature relays packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. - None, unless otherwise specified.

5.3.2 Normal markings. - In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. - The armature relay is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiver Sets AN/BQG-1, -2, -3, -4, or other suitable application.

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6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specifications; applicable drawings and other documents; selections of applicable levels of packaging and packing; number of unit containers per pack; and guarantee clause (see 6.4).

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to para. 4.6.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

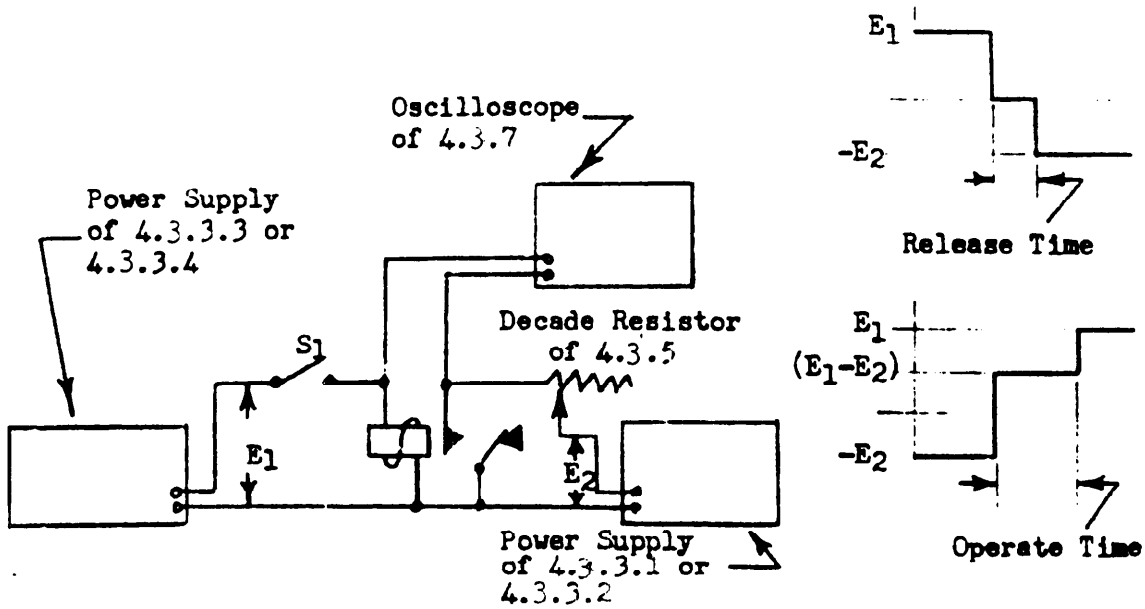
For level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any armature relay which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

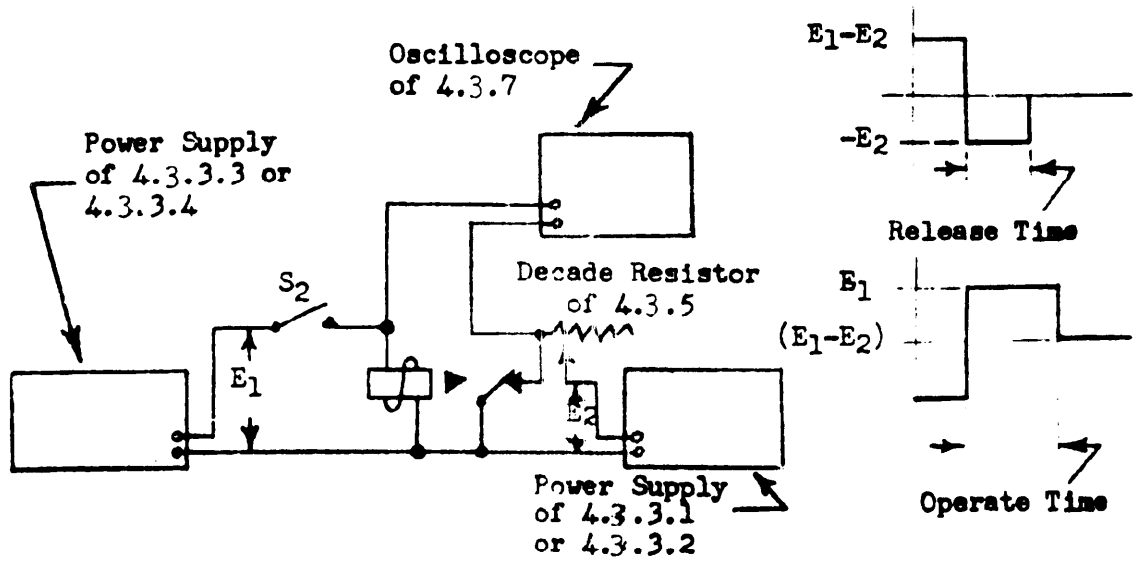
6.5 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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Test circuit and oscilloscope trace for Normally-Open Contacts



Test circuit and oscilloscope trace for Normally-Closed Contacts

Notes:

1. Closing switch S_1 or S_2 produces the oscilloscope trace giving operate time.
2. Opening switch S_1 or S_2 produces the oscilloscope trace giving release time.

Figure 4. Measurement of Operate Time and Release Time

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R800

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO • IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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